THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sting of a bee contains: 1

(a) ethanoic acid **(b)** citric acid **(c)** methanoic acid **(d)** lactic acid

2. The pH of pure water at room temperature is: 1

(a) 5 **(b)** 7 **(c)** 0 **(d)** 14

3. Plaster of Paris should be stored in a moisture-proof container because: 1

(a) it turns blue **(b)** it loses water

(c) it melts **(d)** it absorbs water and sets to gypsum

4. In the chlor-alkali process, the gas liberated at the anode is: 1

(a) carbon dioxide **(b)** oxygen **(c)** chlorine **(d)** hydrogen

5. Acid rain has pH less than: 1

(a) 5.6 **(b)** 7 **(c)** 9 **(d)** 8

6. Which of the following salts does not contain water of crystallisation? 1

(a) gypsum **(b)** washing soda **(c)** baking soda **(d)** blue vitriol

7. Bleaching powder is made by passing chlorine over: 1

(a) quick lime **(b)** dry slaked lime **(c)** limestone **(d)** gypsum

8. Dry HCl gas does not change the colour of dry litmus paper because: 1

(a) it is neutral **(b)** it has no H^+ ions without water

(c) it is a metal **(d)** it is a base

9. Sodium hydrogen carbonate reacts with acetic acid to give a gas which: 1

(a) turns red litmus blue **(b)** burns with a pop

(c) turns lime water milky **(d)** relights a glowing splinter

10. Baking powder is a mixture of baking soda and: 1

(a) bleaching powder **(b)** washing soda **(c)** tartaric acid **(d)** common salt

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do HCl and HNO3 show acidic character whereas alcohol and glucose do not? 2

12. Why is plaster of Paris stored in moisture-proof containers? 2

13. Write the chemical equation for the action of heat on baking soda. 2

14. Why does distilled water not conduct electricity whereas rain water does? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why does tooth decay start when the pH of the mouth falls below 5.5? How does toothpaste help? 3

16. How is washing soda prepared from baking soda? State two uses. 3

17. Why is magnesium hydroxide used as an antacid? Write the reaction. 3

OR

Why do acids not show acidic behaviour in the absence of water? 3

18. What is plaster of Paris? How is it prepared? Write its use. 3

OR

Explain the principle of the soda-acid fire extinguisher. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How does a base react with a metal and with a non-metallic oxide? Give an equation for each. What does this show about the oxides? 5

OR

How are washing soda and plaster of Paris prepared? Give two uses of each. 5

20. Describe the reaction of metal carbonates and hydrogencarbonates with acids. Explain the lime water test, including the effect of excess CO_2. 5

OR

What is water of crystallisation? Describe what happens when blue copper sulphate crystals are heated and then water is added. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND